Data wrangling coding practice

Your name

2025-02-24

0. Load in the `tidyverse` and `openintro` packages in the code chunk below. We will once again work with the `starbucks` data from the `openintro` package.

```
# load packages here
```

1. Wrangle the `starbucks` data to show a data frame that contains the number of each type of Starbucks item. Display the resulting data frame in descending order.

2. Wrangle the data to display the five Starbucks items with the highest amount of calories. Only display these two variables.

3. Three macronutrients serve as sources of calories (i.e. energy) in food: carbohydrates, proteins, and fat. Carbohydrates contain 4 calories per gram, proteins contain 4 calories per gram, and fats contain 9 calories per gram.

Wrangle the data to add a new variable called `theoretical_cals` which represents the number of calories each item in the `starbucks` data theoretically should have based on its levels of carbohydrates, protein, and fat. Store the resulting data frame as a new data frame called `starbucks_new`.

4. Using your new data frame, display a summary table/data frame that shows the standard deviation of the differences between the reported calories and theoretical calories (reported - theoretical). Make sure to specify an informative/nicer name in your summary table.

5. Lastly, make one more summary data frame that shows the same statistics as in number 4, but for each `type` of item.

You should see an interesting standard deviation. Why did that occur? (Hint: look back at your table from Question 1)

Answer:

Once you're finished, be sure to render and submit the PDF file to the corresponding Gradescope assignment!